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Nomor 1

```
print("Titanic Dataset Basic Statistical Data: ")
titanicData.describe().T
```

	count	mean	std	min	25%	50%	75%	max
PassengerId	891.0	446.000000	257.353842	1.00	223.5000	446.0000	668.5	891.0000
Survived	891.0	0.383838	0.486592	0.00	0.0000	0.0000	1.0	1.0000
Pclass	891.0	2.308642	0.836071	1.00	2.0000	3.0000	3.0	3.0000
Age	714.0	29.699118	14.526497	0.42	20.1250	28.0000	38.0	80.0000
SibSp	891.0	0.523008	1.102743	0.00	0.0000	0.0000	1.0	8.0000
Parch	891.0	0.381594	0.806057	0.00	0.0000	0.0000	0.0	6.0000
Fare	891.0	32.204208	49.693429	0.00	7.9104	14.4542	31.0	512.3292

Nomor 2

```
# Soal Nomor 2
femaleSurvChance = titanicData[titanicData["Sex"] ==
"female"]['Survived'].mean() * 100
print("Female Survival Chance: %.0f%%" % femaleAverageSurvChance)
```

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Female Survival Chance: 74%
```

Nomor 3

```
classThreeSurvChance = titanicData[titanicData['Pclass'] ==
3]['Survived'].mean() * 100
print("Third Class Passanger Survival Rate: %0.f%%" % classThreeSurvChance)

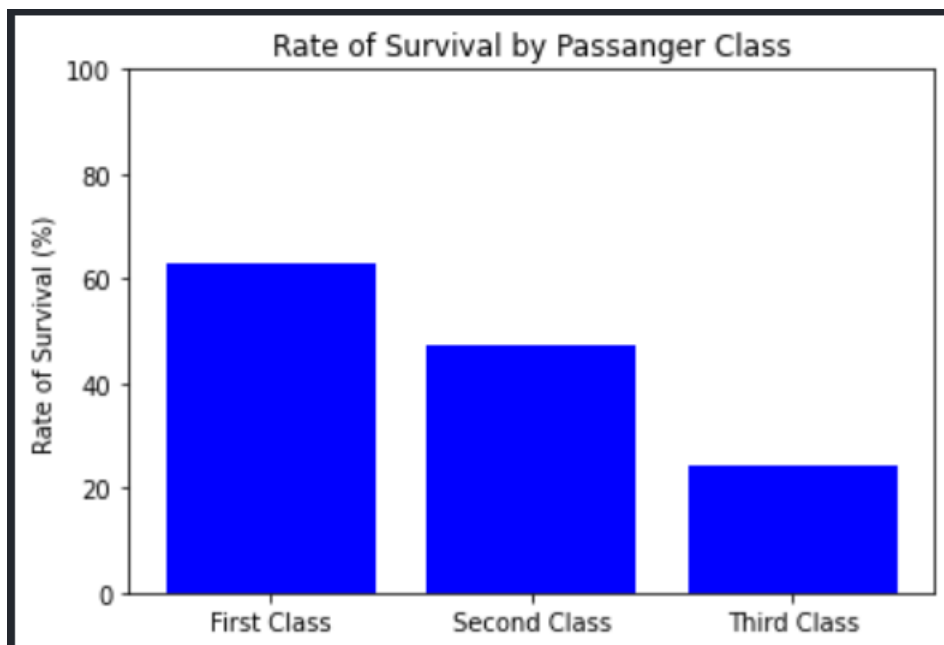
classTwoSurvChance = titanicData[titanicData['Pclass'] ==
2]['Survived'].mean() * 100
print("Second Class Passanger Survival Rate: %0.f%%" % classTwoSurvChance)

classOneSurvChance = titanicData[titanicData['Pclass'] ==
1]['Survived'].mean() * 100
print("First Class Passanger Survival Rate: %0.f%%" % classOneSurvChance)

print("\nSurvival Rate by Passanger Class Bar Distribution")
pClass = ["First Class", "Second Class", "Third Class"]
survRateByClass = [classOneSurvChance, classTwoSurvChance,
classThreeSurvChance]

plt.bar(pClass, survRateByClass, color='blue')
plt.title("Rate of Survival by Passanger Class")
plt.ylabel("Rate of Survival (%)")
plt.ylim(0,100)

plt.show()
```



Nomor 4

```
firstClassFemaleSurvChance = titanicData[(titanicData["Sex"] == "female") &
(titanicData['Pclass'] == 1)]['Survived'].mean() * 100
print("First Class Female Survival Chance: %.0f%%" %
firstClassFemaleSurvChance)
```

First Class Female Survival Chance: 97%

Nomor 5

```
survivedAgeDistribution = list(titanicData[titanicData['Survived'] ==
1]['Age'])
notSurvivedAgeDistribution = list(titanicData[titanicData['Survived'] ==
0]['Age'])

plt.hist([survivedAgeDistribution, notSurvivedAgeDistribution],
label=['Survived', 'Not Survived'], bins=10)

plt.title("Survived and Not Survived Age Distribution")
plt.xlabel('Age')
plt.ylabel('Passangers')
plt.ylim(0,140)
plt.xlim(0,90)
plt.legend()

plt.show()
```

